

Supplementary Table S1. ArcGIS-served 2031 planning objective profiles

Values are means plus or minus population standard deviations across computational seeds 31, 47 and 73. All values are conditional on the ArcGIS-served 2025 origin state and product-specific 2026–2031 scenario actions. Lower values are preferred for all three public proxy objectives. A frontier mark denotes non-domination within that scenario only; it is not a forecast-accuracy or welfare ranking.

Scenario	Model	New built to major roads (m)	New built to prior built (m)	Union-built components per 1,000 cells	Within-scenario frontier
Moderate growth	FLUS-style ANN-CA	368.0 +/- 15.0	194.3 +/- 28.3	0.731 +/- 0.066	*
Moderate growth	Geospatial Kernel	466.9 +/- 5.9	119.1 +/- 1.7	0.990 +/- 0.020	*
Moderate growth	GeoFM-LDN	487.0 +/- 52.9	262.6 +/- 38.0	1.295 +/- 0.059	
Green-priority growth	FLUS-style ANN-CA	366.0 +/- 6.9	184.8 +/- 30.7	0.798 +/- 0.073	*
Green-priority growth	Geospatial Kernel	486.9 +/- 8.7	113.4 +/- 1.0	1.103 +/- 0.018	*
Green-priority growth	GeoFM-LDN	495.1 +/- 59.1	249.4 +/- 36.1	1.320 +/- 0.058	
High outward growth	FLUS-style ANN-CA	388.5 +/- 29.8	228.7 +/- 24.7	0.631 +/- 0.052	*
High outward growth	Geospatial Kernel	447.9 +/- 1.1	155.3 +/- 2.2	0.769 +/- 0.006	*
High outward growth	GeoFM-LDN	465.2 +/- 39.0	295.2 +/- 47.1	1.249 +/- 0.073	

In this product track, FLUS-style ANN-CA has lower mean major-road distance and lower union-built component density, while Kernel has lower mean distance to prior built cells in all three scenarios. This dimension-specific trade-off differs from the Dynamic World-origin table and is why unchanged frontier membership must not be interpreted as unchanged planning behaviour.

Supplementary Table S2. Strict neighbourhood-weight sensitivity

Values are means across seeds 31, 47 and 73 under the destination-correct strict multi-class change Figure of Merit evaluator. This is a post-hoc robustness diagnostic, not a causal estimate of the neighbourhood term.

Target year	Neighbourhood weight λ	Mean strict FoM	Population SD
2023	0.000	0.19570436	0.00171387
2023	0.175	0.19609294	0.00176310
2023	0.350	0.19608163	0.00141267
2023	0.700	0.19577652	0.00133722
2024	0.000	0.25304391	0.00139858
2024	0.175	0.25296735	0.00100364
2024	0.350	0.25294629	0.00083516
2024	0.700	0.25273342	0.00118025

Machine-readable seed-level values are provided in `../benchmarks/abu_dhabi_land_use_v1/artifacts/mechanism_ablations/neighbourhood_weight_sensitivity_report.json`

Supplementary Table S3. FLUS feature-expansion diagnostics

These diagnostics use the author-modified FLUS console with the released public-data benchmark. They are not comparable headline estimators because they retain the FLUS same-year suitability target and cellular-automata allocation semantics rather than the Geospatial Kernel next-state transition and projection contract.

Configuration	Archive / evidence platform	Seeds	2023 strict FoM	2024 strict FoM	2023 predicted changes	2024 predicted changes	2023 demand TV	2024 demand TV	Interpretation
Seven drivers + current- class one-hot (13 features)	macOS arm64	31, 47, 73	0.0000	0.0000	0	0	0.0388	0.0857	Same-year class identity leakage; zero- change diagnostic.
Seven drivers + neighbour- hood fractions (19 features)	macOS arm64	31, 47, 73	0.1071–0.1617	0.0872–0.1386	1,702–2,417	1,805–2,499	0.0153–0.0305	0.0615–0.0767	Partial demand underfill. Observed changes: 4,500 (2023) and 8,464 (2024).
Kernel- order feature family (25 features)	macOS arm64	31, 47, 73	0.0000–0.1320	0.0000–0.1960	0–2,948	0–6,462	0.0023–0.0388	0.0052–0.0857	Seeds 47 and 73 collapsed. Seed 31 is diagnostic only.

Configuration	Archive / evidence platform	Seeds	2023 strict FoM	2024 strict FoM	2023 predicted changes	2024 predicted changes	2023 demand TV	2024 demand TV	Interpretation
Kernel- order feature family (25 features)	independent external Windows x86_64 verification	31, 47, 73	0.0000	0.0000	0	0	0.0388	0.0857	All three seeds collapsed; evidence was reported during anonymous peer review and source rasters are not locally archived.

The 25-feature mode collapsed in five of six platform–seed runs. The sole non-collapsed macOS seed-31 result is not reported as a point estimate or as a Kernel-minus-FLUS paired contrast. FLUS_RANDOM_SEED produces deterministic runs within a fixed platform and build environment, but the console uses C `rand()/srand()` in both ANN sampling and CA roulette draws. The random stream therefore differs across platform runtimes.

The independent external Windows x86_64 verification also found the seven-driver control to differ from the macOS output by approximately 2–4% of valid categorical cells. These external Windows observations are reported here for the platform-boundary diagnosis only. Their source rasters and logs are not part of the repository, so they are not used as author-run primary evidence or as a model-comparison point estimate.

Supplementary Table S4. Dynamic World-to-ArcGIS class matrices

Each matrix uses the pairwise common-coverage 100-m grid. Rows are Dynamic World classes and columns are ArcGIS-served classes after the frozen six-class crosswalk. These are product-label correspondences, not verified land-cover transitions or land-use truth. Zeros are retained explicitly.

2017

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	14887	17	26	37	154	269
Woody	170	97	68	83	21	1
vegetation						
Low vegetation	9	0	297	5	76	1
Wetland	441	148	9	374	1	1
Built	34	0	181	0	8711	179
Bare	4726	3	1630	115	12762	34227

2018

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	14466	17	25	17	148	178
Woody	141	48	48	57	17	0
vegetation						
Low vegetation	7	1	266	2	85	2
Wetland	387	194	7	329	0	1
Built	34	0	135	0	7321	92
Bare	5378	2	1806	79	14163	34316

2019

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	14901	13	26	28	183	183
Woody	182	71	53	112	21	1
vegetation						
Low vegetation	30	1	288	7	102	4
Wetland	412	197	8	390	1	1

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Built	37	0	160	1	8612	97
Bare	4650	6	1798	108	15384	31691

2020

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	15100	14	24	34	190	205
Woody	192	111	71	148	22	0
vegetation						
Low vegetation	30	3	305	32	109	1
Wetland	417	139	7	448	1	2
Built	43	0	139	0	9179	281
Bare	4231	6	1883	183	11928	34283

2021

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	15103	9	33	37	179	173
Woody	188	109	55	170	11	1
vegetation						
Low vegetation	36	5	320	18	88	1
Wetland	434	162	9	407	2	2
Built	34	0	146	0	8249	124
Bare	4222	8	1963	154	13078	34217

2022

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	14930	5	39	29	209	226
Woody	271	42	59	102	21	1
vegetation						

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Low vegetation	45	0	337	0	88	1
Wetland	636	33	22	319	1	0
Built	35	0	157	0	9108	59
Bare	4496	8	2125	40	18787	27534

2023

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	15442	0	39	0	230	276
Woody vegetation	591	9	65	0	23	0
Low vegetation	60	0	313	0	86	2
Wetland	1054	3	20	13	2	4
Built	52	0	155	0	11839	75
Bare	3944	10	1993	11	19108	24331

2024

Dynamic World / ArcGIS-served	Water	Woody	Low vegetation	Wetland	Built	Bare
Water	15570	0	34	0	236	222
Woody vegetation	822	9	59	0	27	0
Low vegetation	32	0	316	0	60	0
Wetland	964	4	12	11	5	4
Built	59	0	193	0	15242	104
Bare	3723	3	1967	9	17987	22075

Supplementary Table S5 | Rolling-origin and external-product diagnostics

The rolling-origin rows use one-step expanding-window Geospatial Kernel fits with observed target class totals supplied only to the oracle-demand allocator. The leakage-controlled fits use 23 features and omit the later road snapshot from every fold. Values are means across seeds 31, 47 and 73; $\pm$ is the sample standard deviation across those three computational seeds. The spatial-block intervals are paired contrasts from 1,000 resamples of 8×8 cells ($800 \text{ m} \times 800 \text{ m}$).

Target year	Kernel strict FoM	Random strict FoM	Persistence strict FoM	Kernel – random 95% interval range across seeds	Kernel – persistence 95% interval range across seeds
2021	0.2595 $\pm$ 0.0046	0.0524 $\pm$ 0.0069	0.0000	[0.1704, 0.2529]	[0.2247, 0.3001]
2022	0.0724 $\pm$ 0.0020	0.0048 $\pm$ 0.0016	0.0000	[0.0543, 0.0812]	[0.0572, 0.0873]
2023	0.2116 $\pm$ 0.0042	0.0200 $\pm$ 0.0005	0.0000	[0.1686, 0.2126]	[0.1873, 0.2334]
2024	0.1816 $\pm$ 0.0022	0.0323 $\pm$ 0.0013	0.0000	[0.1317, 0.1658]	[0.1624, 0.1989]

WorldCover 2020 v1.0 and 2021 v2.0 built-fraction thresholds were 0.05, 0.10, 0.20 and 0.30. The values below are F1 scores for built stock in 2021 and built gain between 2020 and 2021. The original Dynamic World-count gain comparison is structural-zero because Dynamic World built count decreases; the count-controlled rows use the WorldCover gain count as the action and are the informative allocation comparison. All comparisons remain external public-product agreement diagnostics, not authoritative ground-truth validation. For the two count-controlled action columns, $\pm$ is the three-seed sample SD.

Built fraction threshold	WorldCover built gain cells	Dynamic World built-gain F1	Dynamic World 2020 same-year stock F1	Dynamic World 2021 same-year stock F1	Kernel 2021 built-stock F1	Persistence 2021 built-stock F1	Kernel built-gain F1 (structural-zero)	Kernel F1 (WorldCover-count action, mean $\pm$ SD)	Random F1 (WorldCover-count action, mean $\pm$ SD)
0.05	1,802	0.0111	0.3908	0.3501	0.3518	0.3886	0.0000	0.0157 $\pm$ 0.0008	0.0268 $\pm$ 0.0058
0.10	1,566	0.0135	0.3998	0.3603	0.3625	0.4003	0.0000	0.0228 $\pm$ 0.0007	0.0277 $\pm$ 0.0036
0.20	1,445	0.0188	0.4186	0.3780	0.3814	0.4209	0.0000	0.0268 $\pm$ 0.0011	0.0251 $\pm$ 0.0011
0.30	1,511	0.0192	0.4334	0.3924	0.3971	0.4387	0.0000	0.0397 $\pm$ 0.0007	0.0251 $\pm$ 0.0075

Kernel was below random at 0.05, the 0.10–0.20 differences were not distinguishable from three-seed variation, and only the 0.30 result was clearly above random. The low-threshold deficit is consistent with WorldCover cells that barely exceed 5% built fraction, including sparse or road-edge structure that Dynamic World does not consistently label as built.

The rolling 2023 row is a one-step forecast with 23 features and training through 2022, whereas the headline 2023 result uses 25 features and training through 2021. Rolling 2024 is also one-step, while headline 2024 is a two-step open-loop rollout; these rows are not directly comparable point estimates.

Source data and machine-readable details are provided in `artifacts/rolling_backtest/report.json`, `artifacts/rolling_backtest/paired_spatial_uncertainty.json` and `artifacts/external_validation/worldcover/diagnostic.json`.

Supplementary Table S6. Cross-product historical robustness

All values use the frozen Abu Dhabi city 100-m grid. Strict destination-change FoM values are means across seeds 31, 47 and 73. Each expanding-window fold fits only observations available through its origin year; the target label supplies oracle class totals and evaluation only. Dynamic World and the ArcGIS-served Impact Observatory/Microsoft/Esri series retain their product-native label and aggregation-quality semantics, so this is a whole-product-pipeline robustness test rather than an isolated label substitution. Neither product is authoritative local land-use truth.

Product	Target	Observed change (%)	One-year reversion (%)	Random	FLUS-style	Kernel	GeoFM-LDN
Dynamic World	2021	3.05	38.90	0.0492	0.1627	0.2595	0.2469
Dynamic World	2022	3.45	33.31	0.0056	0.0618	0.0724	0.0584
Dynamic World	2023	5.65	10.39	0.0217	0.1150	0.2116	0.1663
Dynamic World	2024	6.24	n/a	0.0312	0.0967	0.1816	0.1661
ArcGIS-served IO/MS/Esri	2021	4.06	42.61	0.0020	0.0625	0.0154	0.0314
ArcGIS-served IO/MS/Esri	2022	10.59	5.53	0.0923	0.2227	0.2775	0.4488
ArcGIS-served IO/MS/Esri	2023	6.07	20.47	0.0471	0.1016	0.1619	0.1561
ArcGIS-served IO/MS/Esri	2024	5.26	28.26	0.0406	0.1197	0.1674	0.1312
ArcGIS-served IO/MS/Esri	2025	4.50	n/a	0.0090	0.0731	0.0579	0.0654

Matched product-difference sensitivity

Intervals are ArcGIS-served minus Dynamic World mean strict FoM for the four paired targets. The 2022 target follows the documented 2021–2022 ArcGIS class-composition break, so the right-hand interval isolates the remaining folds rather than treating its sign as ordinary product variation. A negative value means lower agreement with the ArcGIS-served product under the product-specific pipeline; it does not identify which product is correct.

Model	All matched targets	All-target interval	Excluding 2022	Excluding-2022 interval	Direction excluding 2022
FLUS-style ANN-CA	2021,2022,2023,2024	-0.100 to +0.161	2021,2023,2024	-0.100 to +0.023	mixed remaining folds
Geospatial Kernel	2021,2022,2023,2024	-0.244 to +0.205	2021,2023,2024	-0.244 to -0.014	negative in every remaining fold
GeoFM-LDN	2021,2022,2023,2024	-0.215 to +0.390	2021,2023,2024	-0.215 to -0.010	negative in every remaining fold

Same-year agreement between the two harmonized products is reported below. A 100-m cell contributes only where both products have a usable canonical label.

Year	All-class agreement	Built-class IoU	Dynamic World built (km2)	ArcGIS-served built (km2)
2017	0.7346	0.3938	91.05	217.25
2018	0.7114	0.3328	75.82	217.34
2019	0.7015	0.3501	89.07	243.03
2020	0.7451	0.4193	96.42	214.29
2021	0.7324	0.3765	85.53	216.07
2022	0.6553	0.3200	93.59	282.14
2023	0.6514	0.3750	121.21	312.88
2024	0.6674	0.4494	155.98	335.57

Product-native annual class stocks

Counts below use each product's own city-valid 100-m state layer, rather than the smaller pairwise common-coverage mask. They describe mapped class prevalence, not verified land-cover change or land-use truth.

Product	Year	Valid cells	Water	Woody vegetation	Low vegetation	Wetland	Built	Bare
Dynamic World	2017	79760	15390	440	388	974	9105	53463
Dynamic World	2018	79769	14851	311	363	918	7582	55744
Dynamic World	2019	79759	15334	440	432	1009	8907	53637
Dynamic World	2020	79761	15567	544	480	1014	9642	52514
Dynamic World	2021	79747	15534	534	468	1016	8553	53642
Dynamic World	2022	79765	15438	496	471	1011	9359	52990

Product	Year	Valid cells	Water	Woody vegetation	Low vegetation	Wetland	Built	Bare
Dynamic World	2023	79750	15987	688	461	1096	12121	49397
Dynamic World	2024	79749	16062	917	408	1000	15598	45764
ArcGIS-served IO/MS/Esri	2017	79775	20281	265	2211	615	21725	34678
ArcGIS-served IO/MS/Esri	2018	79775	20419	262	2287	484	21734	34589
ArcGIS-served IO/MS/Esri	2019	79775	20226	288	2333	648	24303	31977
ArcGIS-served IO/MS/Esri	2020	79775	20021	273	2429	850	21429	34773
ArcGIS-served IO/MS/Esri	2021	79775	20035	293	2526	792	21608	34521
ArcGIS-served IO/MS/Esri	2022	79775	20423	88	2739	490	28214	27821
ArcGIS-served IO/MS/Esri	2023	79775	21168	22	2585	24	31288	24688
ArcGIS-served IO/MS/Esri	2024	79775	21193	16	2581	20	33557	22408
ArcGIS-served IO/MS/Esri	2025	79775	21180	2	2694	17	32731	23151

The ArcGIS-served sequence has a material class-composition discontinuity: built rises from 21,608 cells in 2021 to 28,214 in 2022, while bare falls from 34,521 to 27,821; woody vegetation and wetland later approach zero. The upstream Living Atlas item does not publish annual model-version identifiers, so this table cannot assign a cause. It is therefore evidence of a product-series break, not evidence that these mapped transitions occurred on the ground.

The full annual 6 x 6 Dynamic World-to-ArcGIS class matrix is supplied as `results_arcgis_v2/paper_refresh/cross_product_class_matrix.csv`. Its rows use the pairwise common-coverage grid; zeros are retained so every annual matrix is explicit.

Rangeland crosswalk sensitivity

The source service names raw class 11 Rangeland. The frozen benchmark maps it to low vegetation. The counterfactual below maps only raw class 11 to bare before the identical 10 x 10 majority aggregation. No model was refit, so this is a label-aggregation sensitivity, not a historical-skill or planning-rank sensitivity.

Year	Raw rangeland 10-m pixels	Changed 100-m labels	Baseline low vegetation	Alternate low vegetation	Baseline bare	Alternate bare
2017	233306	2265	2211	23	34678	36937
2018	242279	2346	2287	16	34589	36928
2019	247521	2384	2333	16	31977	34355
2020	257835	2484	2429	12	34773	37248
2021	268400	2589	2526	12	34521	37102
2022	289017	2792	2739	24	27821	30605
2023	275034	2652	2585	15	24688	27325
2024	272030	2653	2581	15	22408	25052
2025	282750	2754	2694	21	23151	25891

ArcGIS report SHA-256 (LF-normalized text): d80d2e8ca640686a27cec24cba2ec178dd40ec654ec5f3fd3a63a12c0eb2bcbd.

Dynamic World report SHA-256 (LF-normalized text): bce1043f8478288aff2b5808f3b5521c139c47e265d12a465f928379ff2f7fe8.

Supplementary Table S7. Net-change limits and conditional model contrasts

Derived from frozen three-seed expanding-window reports; no models were refit. Origin counts are recovered from persistence outputs on the same evaluation mask. M is half the L1 difference between origin and feasible target class totals. O is observed gross change. For strict destination-change FoM, hits cannot exceed $\min(M, O)$, and the union contains at least $\max(M, O)$ cells, hence $\text{FoM} \leq \min(M, O) / \max(M, O)$. This loose count-only upper bound is not necessarily attainable: destination, hard-mask and spatial restrictions can lower the achievable maximum. Feasible counts may differ from observed counts. The bound is not a corrected skill score or a validation reference.

The count-only bound applies only to the exact minimum-change projections (Kernel, GeoFM-LDN and random minimum-change), each of which changes exactly M cells. The finite-iteration FLUS-style CA can meet near-exact final class totals through simultaneous class inflows and outflows and therefore can change more than M cells; it is not constrained by this bound. The seed triplets below expose this asymmetric transition budget.

Product	Target	O	M	Exact-count FoM upper bound	FLUS changed cells (31/47/73)	Kernel changed cells (31/47/73)	GeoFM-LDN changed cells (31/47/73)	Random changed cells (31/47/73)	Kernel FoM
Dynamic World	2021	2429	1115	0.4590	5820/5684/4919	1115/1115/1115	1115/1115/1115	1115/1115/1115	0.2595
Dynamic World	2022	2749	793	0.2885	1174/1091/1139	793/793/793	793/793/793	793/793/793	0.0724
Dynamic World	2023	4500	3080	0.6844	3618/3393/4125	3080/3080/3080	3080/3080/3080	3080/3080/3080	0.2116
Dynamic World	2024	4972	3696	0.7434	10219/7017/9558	3696/3696/3696	3696/3696/3696	3696/3696/3696	0.1816
ArcGIS	2021	3241	278	0.0858	6860/779/2291	278/278/278	278/278/278	278/278/278	0.0154
ArcGIS	2022	8446	6864	0.8127	7618/7552/12472	6864/6864/6864	6864/6864/6864	6864/6864/6864	0.2775
ArcGIS	2023	4846	3223	0.6651	5367/3975/4236	3223/3223/3223	3223/3223/3223	3223/3223/3223	0.1619
ArcGIS	2024	4197	2281	0.5435	2822/4781/3080	2281/2281/2281	2281/2281/2281	2281/2281/2281	0.1674
ArcGIS	2025	3587	849	0.2367	1651/957/6693	849/849/849	849/849/849	849/849/849	0.0579

For the ArcGIS folds, the exact-count upper bounds span 0.0858–0.8127 across all targets and 0.0858–0.6651 after excluding the 2022 product-series-break target. This sensitivity concerns the net-versus-gross change relation, not verified land-cover accuracy.

Seed-specific paired spatial-block intervals

Each interval uses the archived 1,000 shared resamples of 8×8 cells. Seeds are computational replicates, not independent field samples. Intervals are conditional on each product, fold, seed and mask; no pooled confidence interval or multiple-comparison adjustment is claimed. An interval containing zero does not establish equivalence. Block-size sensitivity and spatial transfer remain untested.

Product	Target	Contrast	Seed	Median	Lower 95%	Upper 95%	Direction
Dynamic World	2021	Kernel minus FLUS-style	31	0.1057	0.0810	0.1286	positive

Product	Target	Contrast	Seed	Median	Lower 95%	Upper 95%	Direction
Dynamic World	2021	Kernel minus FLUS-style	47	0.0993	0.0764	0.1235	positive
Dynamic World	2021	Kernel minus FLUS-style	73	0.0841	0.0616	0.1090	positive
Dynamic World	2021	GeoFM-LDN minus Kernel	31	-0.0249	-0.0456	-0.0044	negative
Dynamic World	2021	GeoFM-LDN minus Kernel	47	-0.0028	-0.0210	0.0177	includes_zero
Dynamic World	2021	GeoFM-LDN minus Kernel	73	-0.0103	-0.0290	0.0077	includes_zero
Dynamic World	2022	Kernel minus FLUS-style	31	0.0088	-0.0010	0.0180	includes_zero
Dynamic World	2022	Kernel minus FLUS-style	47	0.0137	0.0037	0.0248	positive
Dynamic World	2022	Kernel minus FLUS-style	73	0.0093	0.0002	0.0196	positive
Dynamic World	2022	GeoFM-LDN minus Kernel	31	-0.0178	-0.0279	-0.0079	negative
Dynamic World	2022	GeoFM-LDN minus Kernel	47	-0.0131	-0.0245	-0.0022	negative
Dynamic World	2022	GeoFM-LDN minus Kernel	73	-0.0109	-0.0214	-0.0014	negative
Dynamic World	2023	Kernel minus FLUS-style	31	0.0987	0.0844	0.1138	positive
Dynamic World	2023	Kernel minus FLUS-style	47	0.0488	0.0375	0.0599	positive
Dynamic World	2023	Kernel minus FLUS-style	73	0.1411	0.1253	0.1592	positive
Dynamic World	2023	GeoFM-LDN minus Kernel	31	-0.0420	-0.0568	-0.0272	negative
Dynamic World	2023	GeoFM-LDN minus Kernel	47	-0.0449	-0.0613	-0.0289	negative
Dynamic World	2023	GeoFM-LDN minus Kernel	73	-0.0478	-0.0637	-0.0318	negative
Dynamic World	2024	Kernel minus FLUS-style	31	0.1107	0.0964	0.1267	positive
Dynamic World	2024	Kernel minus FLUS-style	47	0.0806	0.0602	0.0997	positive
Dynamic World	2024	Kernel minus FLUS-style	73	0.0627	0.0491	0.0773	positive

Product	Target	Contrast	Seed	Median	Lower 95%	Upper 95%	Direction
Dynamic World	2024	GeoFM-LDN minus Kernel	31	-0.0170	-0.0280	-0.0053	negative
Dynamic World	2024	GeoFM-LDN minus Kernel	47	-0.0095	-0.0213	0.0030	includes_zero
Dynamic World	2024	GeoFM-LDN minus Kernel	73	-0.0197	-0.0320	-0.0078	negative
ArcGIS	2021	Kernel minus FLUS-style	31	-0.0698	-0.0866	-0.0543	negative
ArcGIS	2021	Kernel minus FLUS-style	47	-0.0179	-0.0282	-0.0059	negative
ArcGIS	2021	Kernel minus FLUS-style	73	-0.0536	-0.0671	-0.0411	negative
ArcGIS	2021	GeoFM-LDN minus Kernel	31	0.0173	0.0031	0.0371	positive
ArcGIS	2021	GeoFM-LDN minus Kernel	47	0.0157	0.0001	0.0399	positive
ArcGIS	2021	GeoFM-LDN minus Kernel	73	0.0131	-0.0010	0.0306	includes_zero
ArcGIS	2022	Kernel minus FLUS-style	31	0.1537	0.1269	0.1799	positive
ArcGIS	2022	Kernel minus FLUS-style	47	-0.0145	-0.0487	0.0181	includes_zero
ArcGIS	2022	Kernel minus FLUS-style	73	0.0257	-0.0134	0.0669	includes_zero
ArcGIS	2022	GeoFM-LDN minus Kernel	31	0.1588	0.1266	0.1947	positive
ArcGIS	2022	GeoFM-LDN minus Kernel	47	0.1743	0.1391	0.2084	positive
ArcGIS	2022	GeoFM-LDN minus Kernel	73	0.1783	0.1418	0.2121	positive
ArcGIS	2023	Kernel minus FLUS-style	31	0.0596	0.0429	0.0761	positive
ArcGIS	2023	Kernel minus FLUS-style	47	0.0451	0.0315	0.0590	positive
ArcGIS	2023	Kernel minus FLUS-style	73	0.0759	0.0495	0.1001	positive
ArcGIS	2023	GeoFM-LDN minus Kernel	31	-0.0020	-0.0222	0.0182	includes_zero
ArcGIS	2023	GeoFM-LDN minus Kernel	47	-0.0102	-0.0299	0.0078	includes_zero

Product	Target	Contrast	Seed	Median	Lower 95%	Upper 95%	Direction
ArcGIS	2023	GeoFM-LDN minus Kernel	73	-0.0049	-0.0249	0.0144	includes_zero
ArcGIS	2024	Kernel minus FLUS-style	31	0.0502	0.0299	0.0722	positive
ArcGIS	2024	Kernel minus FLUS-style	47	0.0435	0.0209	0.0637	positive
ArcGIS	2024	Kernel minus FLUS-style	73	0.0492	0.0286	0.0707	positive
ArcGIS	2024	GeoFM-LDN minus Kernel	31	-0.0461	-0.0705	-0.0215	negative
ArcGIS	2024	GeoFM-LDN minus Kernel	47	-0.0125	-0.0407	0.0127	includes_zero
ArcGIS	2024	GeoFM-LDN minus Kernel	73	-0.0495	-0.0756	-0.0233	negative
ArcGIS	2025	Kernel minus FLUS-style	31	-0.0053	-0.0173	0.0073	includes_zero
ArcGIS	2025	Kernel minus FLUS-style	47	0.0095	-0.0054	0.0262	includes_zero
ArcGIS	2025	Kernel minus FLUS-style	73	-0.0492	-0.0669	-0.0323	negative
ArcGIS	2025	GeoFM-LDN minus Kernel	31	0.0020	-0.0143	0.0189	includes_zero
ArcGIS	2025	GeoFM-LDN minus Kernel	47	0.0020	-0.0130	0.0174	includes_zero
ArcGIS	2025	GeoFM-LDN minus Kernel	73	0.0172	0.0003	0.0362	positive